Points: 5.25/19
(27.63 %)

Aufgabe 1.)

Let $\mathbb{Q} \subseteq K$ be a finite galois extension
Let $I, J \subseteq \mathbb{Z}_K$ be ideals and $\sigma \in Gal(K/\mathbb{Q})$

i) Let $a \in \mathbb{Z}_K$.

t.s. $\sigma(a) \in \mathbb{Z}_K$.

proof By assumption $a \in K$ and there exists a monic $P \in \mathbb{Z}[x]$ s.t. $P(a) = 0$. Since $\mathbb{Z} \subseteq \mathbb{Q}$ it follows that $P = \min_{K/\mathbb{Q}}(a)$. Thm by algebra we know that $\sigma(a)$ must be a root of P as well. it follows that $\sigma(a) \in \mathbb{Z}_K$. (lazy! you could have shown that) (1.75)

ii)

t.s. $\sigma(I) \subseteq \mathbb{Z}_K$ ideal.

proof Let $\tilde{a}, \tilde{b} \in \sigma(I)$ Thm. $\exists a, b \in I$ s.t. $\sigma(a) = \tilde{a}$, $\sigma(b) = \tilde{b}$. ok
Thm. since I is an ideal we get by the homomorphism of σ that.

$$\sigma(a) - \sigma(b) \overset{\checkmark}{=} \sigma(\underbrace{a-b}_{\in I}) \in \sigma(I) \quad \checkmark$$

Thm let $r \in \mathbb{Z}_K$. Thm again we get

$$r\sigma(a) \overset{\text{False!}}{=} \sigma(\underbrace{ra}_{\in I}) \in \sigma(I)$$

σ is a $\mathbb{Q}$-homomorphism and $\mathbb{Z}_K \cap \mathbb{Q} = \mathbb{Z}$

Furthermore by (i) we get that $\sigma(I) \subseteq \mathbb{Z}_K$ which implies that $r(I) \subseteq \mathbb{Z}_K$ ideal. (1)

(ii)

t.s. if I prime ideal $\Rightarrow \sigma[I]$ prime ideal.

proof Let $a, b \in \mathbb{Z}_K$ s.t. $\sigma(a \cdot b) \in \sigma(I)$.
Thm $\exists c \in I$ s.t. $a \cdot b = \sigma^{-1}(c)$ and furthermore there exist $a', b' \in I$ s.t. $c = a' \cdot b'$. (vague & confused) By assumption it holds that $a \in I$ or $b \in I \Rightarrow \sigma(a) \in \sigma(I)$ or $\sigma(b) \in \sigma(I)$
w.l.o.g we may assume $\sigma(a) \in \sigma(I)$. Thm it holds that

$$\sigma(a) \cdot \sigma(b) = \sigma(a \cdot b) \overset{?}{=} a \cdot b$$

(σ an $\mathbb{Q}$-algebra)

Since $\mathbb{Z}_K$ is noetherian, we can find a prime $p \in I$ s.t. $p \mid a$??

???

Thm
$\sigma(p) \mid \sigma(a)$ thus $\sigma(p) \mid a \cdot b$
Since σ is a automorphism, it holds that $\sigma(p)$ is prime as well
thus $\sigma(p) \mid a$ or $\sigma(p) \mid b \Rightarrow a \in \sigma(I)$ or $b \in \sigma(I)$
therefore $\sigma(I)$ is prime. (0.5)

(iv)

t.s. $\sigma(IJ) = \sigma(I)\,\sigma(J)$

proof Let $a \in \sigma(IJ)$.

Th. $a = \sigma\left(\sum_i a_{i1} \cdot a_{i2}\right)$ $a_{i1} \in I$, $a_{i2} \in J$. ok

By homomorphism.

$$a = \sum_i \underbrace{\sigma(a_{i1})}_{\in \sigma(I)} \underbrace{\sigma(a_{i2})}_{\in \sigma(J)} \in \sigma(I)\cdot\sigma(J)$$ since $\sigma(I)\cdot\sigma(J)$ a ring.

thus. $\sigma(IJ) \subseteq \sigma(I)\sigma(J)$.

On the contrary. let

$a \in \sigma(I)\sigma(J)$.

Th $a = \sum_i \sigma(a_{i1})\sigma(a_{i2})$ $a_{i1,2} \in I$, $a_{i2} \in J$. ok

hom. $\Rightarrow a = \sigma\left(\underbrace{\sum_i a_i a_{i2}}_{\in I}\right) \in \sigma(IJ)$

and the equality follows.

(2)